Name: ANSWERS. Date: ________

Year 9 Mathematics – ACE and Maritime

Test 1, 2015

Topic –Rates, Graphs, Indices

SAFETY BAY SENIOR HIGH SCHOOL
imagine believe achieve

Rockingham Senior High School

71

=_____ %

Total Time:	*60 minutes*
Total Reading:	*5 minutes*
Total Working:	*55 minutes*
Weighting:	_____*% of the year.*

Equipment:
½ page notes (A4 one side), Scientific Calculator

SECTION 1: CALCULATOR FREE

Time:	*38 minutes*	**Marks for Section 1:**	*36*
Reading:	*2 minutes*	**Equipment Allowed:**	*Nil*
Working:	*36 minutes*		

1. **[4 marks: 1, 1, 1, 1]**
Simplify each of the following

a) $a^3 \times a^6$

$= a^{3+6}$
$= a^9$

b) $5m^4 \times 7m^5$

$= 35m^{4+5}$
$= 35m^9$

c) $2xy^5 \times x^3y^2$

$= 2x^4y^7$

d) $8d^7e^3 \times d^2 \times 3de$

$= 24d^{7+2+1}e^{3+1}$
$= 24d^{10}e^4$

2. **[4 marks: 1, 1, 1, 1]**
Simplify each of the following

a) $\dfrac{y^6}{y^2}$ $= y^{6-2} = y^4$

b) $\dfrac{21a^8}{3a^6}$ $= 7a^{8-6}$
$= 7a^2$

c) $\dfrac{4c^{11}}{24c^4}$ $= \dfrac{1c^{11-4}}{6}$
$= \dfrac{c^7}{6}$

d) $\dfrac{36k^{15}}{15k^8}$
$= \dfrac{12k^{15-8}}{5}$
$= \dfrac{12k^7}{5}$

3. **[8 marks: 2, 2, 1, 3]**
Simplify each of the following

a) $\frac{4p^3 \times 8p^6}{2p^2}$

$= \frac{32p^{3+6}}{2p^2}$

$= \frac{32p^9}{2p^2} = 16p^{9-2}$

$= 16p^7$

b) $\frac{3f^4 \times 15g^6}{5f \times g^2}$

$= \frac{45f^4g^6}{5fg^2}$

$= 9f^{4-1}g^{6-2}$

$= 9f^3g^4$

c) $\frac{40m^9n^{14}}{8m^6n^8}$

$= 5m^{9-6}n^{14-8}$

$= 5m^3n^6$

d) $\frac{3a^3 \times 4b^5 \times c^7}{5a \times b^5 \times 6c^4}$

$= \frac{12a^3b^5c^7}{30ab^5c^4}$ ✓

$= \frac{2a^{3-1}b^{5-5}c^{7-4}}{5}$

$= \frac{2a^2b^0c^3}{5}$

$= \frac{2a^2c^3}{5}$

4. **[4 marks: 1, 1, 1, 1]**
Simplify each of the following

a) $(f^3)^{11}$

$= f^{33}$

b) $(m^5n^2)^6$

$= m^{30}n^{12}$

c) $(3x^4y^3)^3$

$= 3^3x^{12}y^9$

$= 27x^{12}y^9$

d) $(p^3)^2 \times (q^5)^4$

$= p^6 \times q^{20}$

5. [4 marks: 2, 2]
Simplify each of the following

a) $\left(\frac{2d^5}{e^7}\right)^3$

$= \frac{2^3 d^{15}}{e^{21}}$

$= \frac{8d^{15}}{e^{21}}$

b) $\left(\frac{-2a^2b^9}{ab^7}\right)^4$

$= \frac{(-2)^4 a^8 b^{36}}{a^4 b^{28}}$

$= 16a^{8-4}b^{36-28}$

$= 16a^4b^8$

6. [2 marks: 1, 1]
Write each of the following using a negative index

a) $\frac{1}{7}$

$= 7^{-1}$

b) $\frac{1}{x}$

$= x^{-1}$

7. [5 marks: 1, 1, 1, 2]
Use index laws to simplify each of the following, express your answer with **positive indices.**

a) $p^2 \times p^{-5}$

$= p^{2+(-5)}$

$= p^{-3}$

$= \frac{1}{p^3}$

b) $\frac{24w^3}{16w^7}$

$= \frac{3w^{3-7}}{2}$

$= \frac{3w^{-4}}{2}$

$= \frac{3}{2w^4}$

c) $7a^4b \times 3a^{-6}b^3$

$= 21a^{4+(-6)}b^{1+3}$

$= 21a^{-2}b^4$

$= \frac{21b^4}{a^2}$

d) $18x^{-4}y^9 \div 9x^2y^3$

$= 2x^{-4-2}y^{9-3}$

$= 2x^{-6}y^6$

$= \frac{2y^6}{x^6}$

8. **[5 marks: 1, 1, 1, 2]**
Use index laws to simplify each of the following, express your answer with **positive indices.**

a) $\sqrt{9m^2}$

$= \sqrt{9} \times \sqrt{m^2}$
$= 3 \times (m^2)^{1/2}$
$= 3m$

b) $\sqrt{49t^4}$

$= \sqrt{49} \times \sqrt{t^4}$
$= 7 \times (t^4)^{1/2}$
$= 7t^2$

c) $\sqrt[3]{x^3}$

$= (x^3)^{1/3}$
$= x$

d) $\sqrt[3]{125d^6e^3}$

$= \sqrt[3]{125} \times \sqrt[3]{d^6} \times \sqrt[3]{e^3}$
$= 5 \times (d^6)^{1/3} \times (e^3)^{1/3}$

$= 5d^2e$

~ END OF TEST SECTION 1 ~

Name: ______________________________ Date: __________

SECTION 2: CALCULATOR ASSUMED

Time:	*27 minutes*	**Marks for Section 2:**	*35*
Reading:	*1 minutes*	**Equipment Allowed:**	*½ page notes (A4 one side), Scientific calculator*
Working:	*26 minutes*		

9. **[5 marks: 2, 2, 1]**
y is directly proportional to x, and $y = 24$ when $x = 6$.

a) Find k, the constant of proportionality and the rule relating x and y

$y = kx$

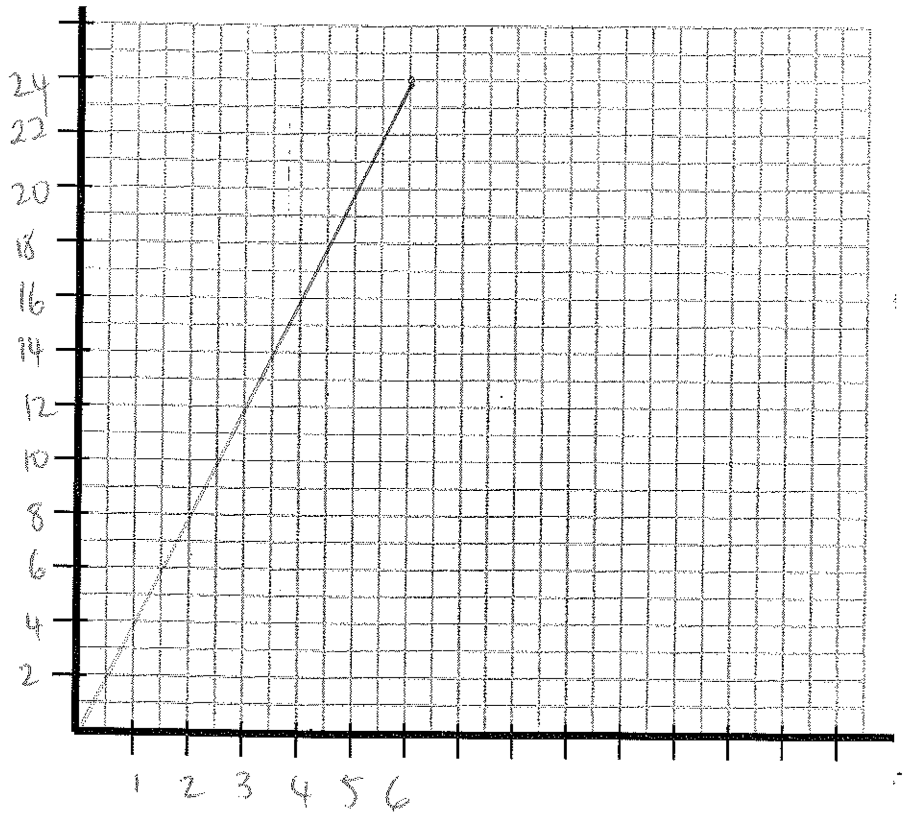
$k = \frac{y}{x}$

$k = \frac{24}{6}$

$= 4$

$\therefore y = 4x$

b) Plot the graph relating x and y. (Hint: you may want to create a table of values)



c) Find the value of y when $= 14$.

$y = 4x$

$y = 4 \times 14$

$= 56$

10. **[4 marks: 1, 2, 1]**

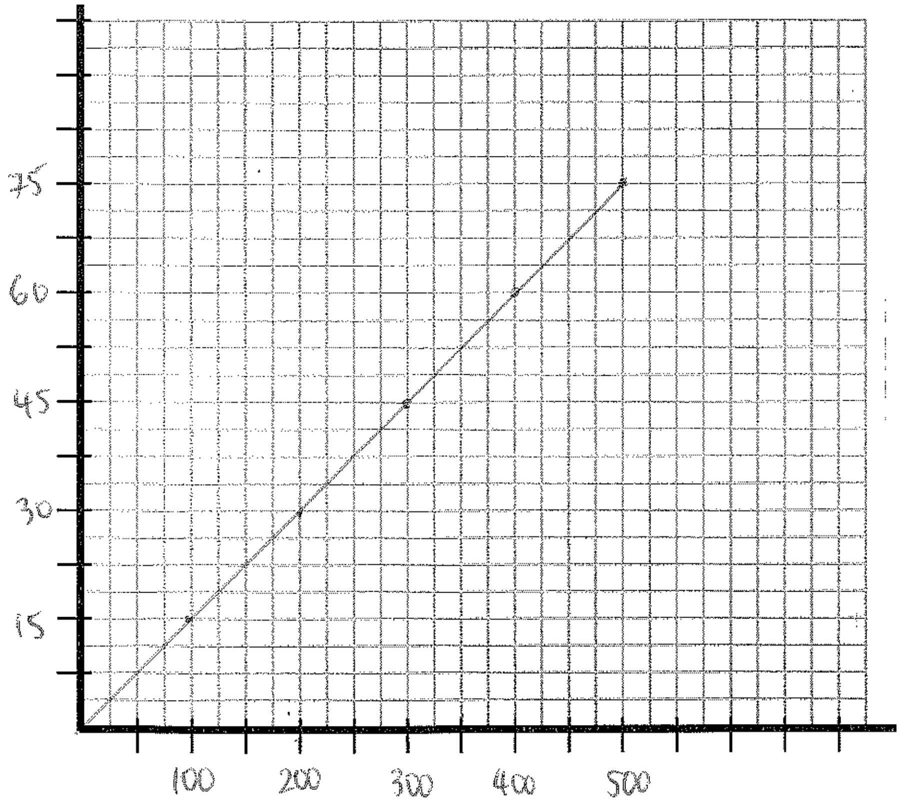
An employee gets \$0.15 for each leaflet delivered. Use x as the number of leaflets delivered and y as the total money they earn in a week.

a) Determine the rule relating y and x

$y = 0.15x$

x	100	200	300	400	500
y	15	30	45	60	75

b) Plot the graph relating y and x (for values of x in units of 100)



c) Determine how much money the person earns if they deliver 1245 leaflets.

$y = 0.15 \times 1245$

$= \$186.75$

11. **[4 marks: 2, 2]**
Find k, the constant of proportionality if:

a) a is directly proportional to b, and $a = 275$ when $b = 25$

$$k = \frac{a}{b}$$
$$= \frac{275}{25}$$
$$= 11$$

b) m is directly proportional to n, and $m = 42$ when $n = 28$

$$k = \frac{m}{n}$$
$$= \frac{42}{28}$$
$$= \frac{3}{2}$$

12. **[4 marks: 2, 1, 1]**
8 calculators cost \$116. Let c = cost of calculator, and let n = number of calculators

a) Write a rule for the relationship between c and n

$\$116 \div 8 = \$14 \cdot 50$

$C = \$14 \cdot 50n$

b) How much would 10 calculators cost in total?

$$C = 14 \cdot 50 \times 10$$
$$= \$145$$

c) How many calculators could I buy for \$200

$$C = 14 \cdot 5n$$
$$200 = 14 \cdot 5n$$
$$\frac{200}{14 \cdot 5} = n$$

$n = 13$ calculators.

13. [4 marks: 2, 2]

a varies inversely as b , and $a = 8$ when $b = 10$

a) Calculate k, the constant of proportionality and the rule relating a and b

$y = \frac{k}{x}$

$\therefore k = x \times y$

$k = a \times b$

$= 8 \times 10$

$= 80$

$\therefore a = \frac{80}{b}$

b) Find b when $a = 17$, correct to 2 decimal places.

$a = \frac{80}{b}$

$17 = \frac{80}{b}$

$17 \times b = 80$

$b = \frac{80}{17}$

$= 4 \cdot 71$ (2 d.p)

14. [5 marks: 2, 2, 1]

y is inversely proportional to x, and $y = 4$ when $x = 6$.

a) Find k, the constant of proportionality and the rule relating y and x

$k = x \times y$

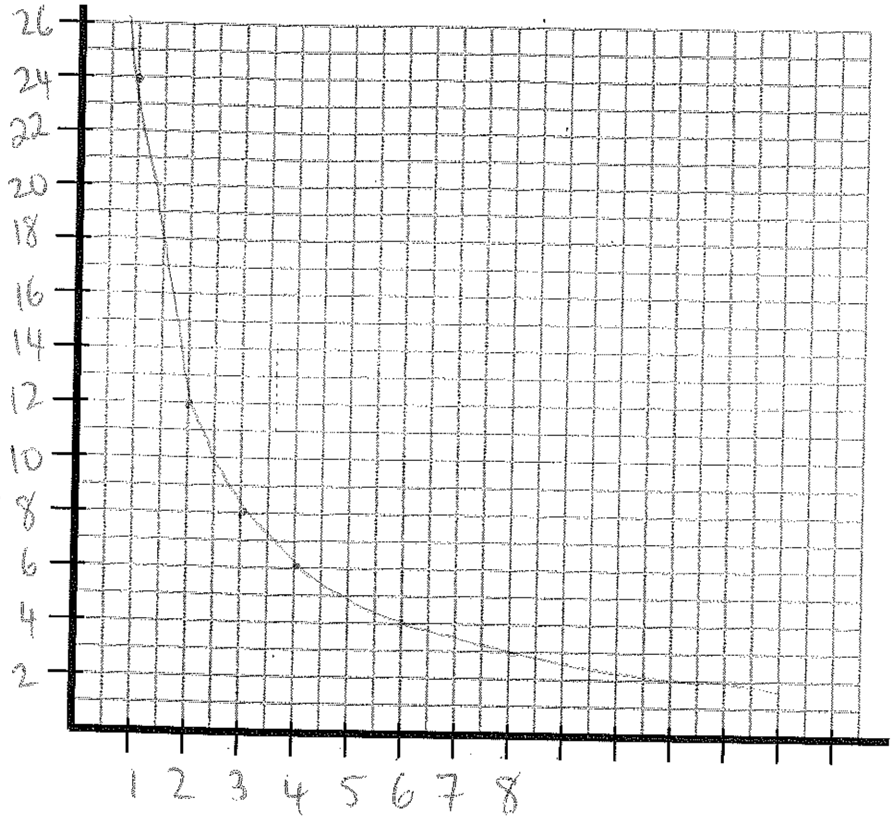
$k = 4 \times 6$

$= 24$

$\therefore y = \frac{24}{x}$

b) Plot the graph relating y and x (for values of x that are factors of k up to 12)

$y = \frac{24}{x}$



x	1	2	3	4	6	8
y	24	12	8	6	4	3

c) Find the value of y when $x = 10$

$y = \frac{24}{x}$

$y = \frac{24}{10} = 2.4$

15. [5 marks: 2, 2, 1]

The table below shows the distance travelled, D, as a person runs for T seconds.

***T* (seconds)**	1.85	3.78	9.69
***D* (meters)**	10	30	100

a) Find the rate of metres/second between the time 1.85 seconds and 3.78 seconds

$\text{Rate} = \frac{\text{meters}}{\text{sec}}$

$= \frac{30-10}{3.78-1.85} = \frac{20}{1.93} = 10.36 \text{ m/sec}$

b) Find the rate metres/second between the distance 30 and 100m.

$\text{Rate} = \frac{100-30}{9.69-3.78}$

$= \frac{70}{5.91} = 11.84 \text{ m/sec}$

c) Is the person's speed constant or variable? Why?

Person's speed is variable as rate is changing over time.

$Rate = \frac{y}{x} \quad \therefore \ x = \frac{y}{Rate} = \frac{y}{2 \cdot 2}$

16. [2 marks]
A car uses petrol at a rate of 2.2L/km. Fill in the table below, correct to 2 decimal places.

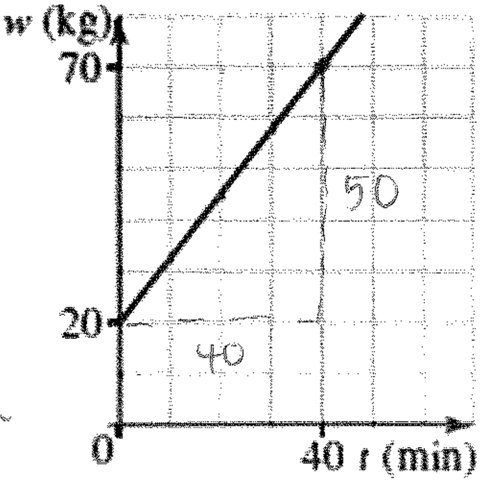
x= distance (km)	0	4·55	6·82	18·18	22·73	27·27
y= petrol (L)	0	10	15	40	50	60

17. [2 marks]
The graph shows the partial variation between mass, w kg, of a tank of water t minutes after the water is poured in it. Find the rate of change of mass in kg/min.

$$\text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{50}{40} = 1 \cdot 25$$

gradient = Rate

$\therefore$ Rate = 1·25 kg/min



~ **END OF TEST SECTION 2** ~